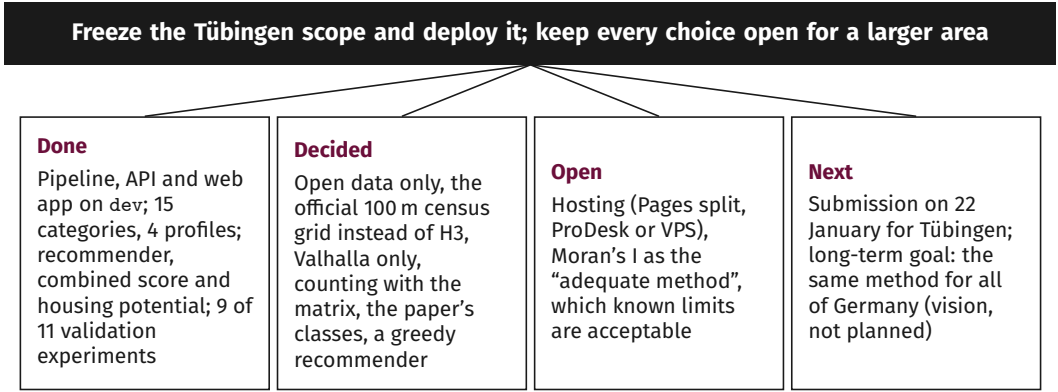
reach: our decisions, open questions and the way to all of Germany

Supervisor meeting · DS500 Data Science Project, Topic 9 “The 15-Minute City”

Rico · Socke · Juju
University of Tübingen, winter term 2026/27
12 November 2026

reach runs end to end for Tübingen; today we explain the decisions behind it, ask four questions and sketch the way to all of Germany



Source: run 20261110T233930Z (OSM 8 Nov 2026); validation report (run 20261106T142630Z); docs/architecture.md §13; docs/deployment_proposal.md (proposal, PR #64).

dev now measures 15 categories for every resident of Tübingen and publishes them in one validated run

1,699
populated 100 m cells (90,355 residents)

15 × 4
categories × profiles

1,238
facilities in the routing area

224
recommended zones; 830 housing cells

Merged into `dev` since 10 November

- #57: parks and playgrounds separate; private ones no longer count
- #58: combined score 0–100 over chosen categories (display only)
- #59: housing potential for cells without residents
- #60: eight extended categories in six groups
- #64: hosting proposal (document only, not approved)

All 15 result checks pass; three warn.

Source: run 20261110T233930Z (OSM 8 Nov 2026), validate_results report part 2; git log of `dev`; docs/features.md.

Every layer of reach rests on a deliberate choice; for each we can name the alternative we rejected

Layer	Our choice	Instead of
Data	Open data, one snapshot per run	Public APIs at runtime (Overpass, Nominatim)
Spatial unit	Official Zensus 2022 100 m grid, cell centroid	H3 hexagons, building origins
Study area	Every resident of the city, 4 km routing buffer	Built-up area only
Routing	Valhalla, self-hosted, one tile set	OSMnx/NetworkX, pandana, r5py
Counting	Time matrix; isochrones only drawn	Point-in-polygon with isochrones
Profiles	Four profiles, 3.39 km/h default, slopes on	One average speed, flat ground
Scoring	k_c, three classes, deficit (Mouratidis 2026)	One stored composite index
Recommender	Greedy on the deficit, 100 m zones	Exact solver in every run, exact sites
System, team	One command, validated publish, docs first	Manual steps, notebooks

On the next slides, ID = recorded decision (O: architecture, M: methodology, W: app iteration, T: team, N: notes); *implicit* = follows from a design principle, no ID.

Source: docs/architecture.md §2, §12, §13; methodology.md; notes.md; AGENTS.md (architecture rules).

We use only open data and read it from one snapshot per run, so every number can be traced and reproduced

ID	Decision	Instead of	Why
implicit	One Geofabrik OSM extract feeds street network, facilities, boundaries and land use	Overpass, Nominatim at runtime	No rate limits; facilities match the street graph
implicit	One snapshot per run with dates and checksums (run id)	Always the latest data	Every result traces back to its inputs
implicit	Demand = Zensus 2022 residents per 100 m cell	Modelled population	Official, open licence, nationwide
O18	Transit = scheduled departures on a reference Tuesday (GTFS), no transit routing	Valhalla multimodal, r5py	Cheap and robust; routing is outside Core
V2	The city's facility lists only validate OSM	Merge them into the counts	No licence stated

Known data limits: census values of small cells are perturbed (cell-key method), main residences only, May 2022.

Source: docs/architecture.md §2 (principle 3), §4, §12, O18; AGENTS.md rule 8; validation report V2.

We keep the official Zensus 100 m grid instead of H3 hexagons, so every resident count stays exactly as published

	Zensus 100 m grid (chosen)	H3 hexagons (rejected)
Residents	Exact counts as published	Re-gridding spreads residents by area; errors at every edge
Identity	Official `GITTER_ID` (EPSG:3035) joins other census grid tables (age: M6)	Own index; every join needs re-gridding
Scale	Same grid for all of Germany: 3,088,037 populated cells in the file we already download	Global, but residents must be disaggregated first
Trade-off	Squares: 8 neighbours at two distances	Hexagons: 6 equal neighbours, nicer for spatial statistics

Same grid everywhere: scored cells, recommender candidates, housing cells and the pin-drop snap to it. Origin = cell centroid (O5); with building origins only 1.9 % of residents × categories change class (V6, walking).

Source: docs/architecture.md §2 (principle 4), §5.2, §12 (H3 aggregation out of scope), O5; validation report V6 (run 20261106T142630Z); Zensus file of run 20261110T233930Z.

We score every resident of the city, but draw the city and place recommendations only where it is built up

ID	Decision	Instead of	Why
O11	Score all populated cells whose centroid lies in the city; `validate` checks $\geq$ 98 % coverage	Built-up area only	The land-use boundary left 8 % of residents unscored (N3)
O17	Urban boundary from residential, commercial, retail, education, institutional and religious land use, $\pm$100 m	Including industrial	Industrial joins separate districts (Bühl–Kilchberg)
O13, O20	Routing area = city + 4 km; the prefilter never exceeds it	City boundary only	Shops in neighbouring towns count; cycling reaches 3.75 km
implicit	Study area = OSM admin level + official municipality key (AGS) in the config	A drawn polygon	Another municipality is a config change

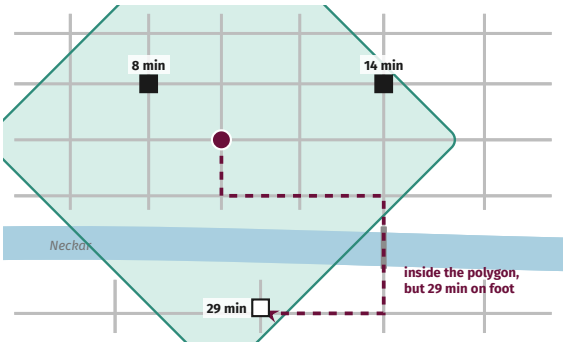
Source: docs/architecture.md §4, §13 (O11, O13, O17, O20), change log v0.12; notes N3; config/config.yaml (`study_area`).

Valhalla is our only routing engine, self-hosted, so batch numbers and live answers always agree

ID	Decision	Instead of	Why
implicit	Valhalla for matrix, isochrone, route and street spread	OSMnx/NetworkX, pandana, r5py	One graph and one snapshot: map and pin-drop give the same numbers
O3	Valhalla runs permanently on our server, internal only	Public routing APIs	No rate limits; nothing leaves the server
implicit	One tile set; the costing is chosen per request	One graph per profile	A new profile needs no rebuild
N8	`costmatrix` for every matrix request	Valhalla's automatic choice	It made 2.5 % of live and stored pairs differ
O4, O19	One-shot tile build on the host, atomic swap	Rebuild inside the live service	Weekly refresh without downtime

Source: AGENTS.md (architecture rule 1); docs/architecture.md §2, §6.1, §6.5, §7, §13; notes N8 (architecture change log v0.15).

We count facilities along the network with a time matrix; isochrone polygons would add 18 % unreachable pairs



D2: for each block of cells one Valhalla matrix to all facilities in reach; a facility counts if its time is $\leq$ 15 min. Isochrones are only drawn.

V1 (150 cells, walking 3.39 km/h): polygons add 18 % unreachable pairs, 47 % of them across the Neckar, rail lines or main roads.

O20: an exact straight-line prefilter; 0 of 1,200 sampled pairs outside the counts are reachable by `/route`.

***implicit*:** raw counts and times are stored; classes are derived at read time, so a new k_c needs no re-routing.

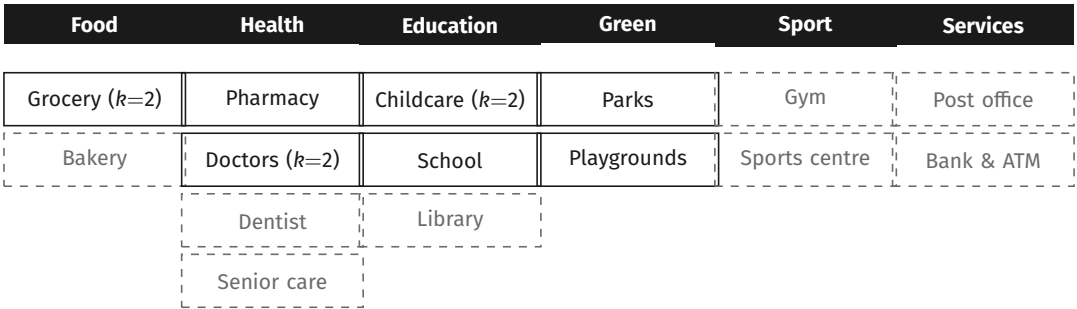
Source: docs/architecture.md §6.3 (D2), §6.5, O20; validation report V1 (run 20261106T142630Z); run 20261110T233930Z (`false_negatives`).

We measure four ways of travelling, with a slow inclusive walking speed and Valhalla's slope model switched on

ID	Decision	Instead of	Why
O6	Walking 3.39 km/h as default (850 m in 15 min), 5.0 km/h for comparison	One average speed	Inclusive; shares change strongly with speed (V5)
O6, O15	Cycling at 15 km/h (hybrid bike)	–	Its 3.75 km reach sets the routing buffer
O6	Wheelchair = pedestrian costing without steps	–	Known limit: steep sections are not avoided (N11)
O7	Valhalla's slope model kept for every profile	Flat-ground times	Tübingen is hilly; slopes change 3.3–3.6 % of classes (V11)
M	15 min to facilities, 10 min on foot to transit stops	–	Topic of the course; methodology §1

Source: docs/architecture.md §6.1, §13 (O6, O7, O15); methodology.md §1; notes N5, N11; validation report V5, V11 (run 20261106T142630Z).

Fifteen categories in six groups: seven Core ones follow the paper, eight extended ones are indicative



- **O8:** coarse categories keep OSM gaps small
- **W7a:** private parks and playgrounds dropped (68)
- **N16:** restaurants dropped; the check was kept
- Dashed: unchecked; $k = 1$ unless stated

Source: docs/methodology.md §2 (v0.11), §3; notes N14, N16; architecture.md O8; run 20261110T233930Z.

Each cell gets a class per category and a deficit weighted by residents; the combined score is for display only

Classes (Mouratidis 2026)

Lack	$A = 0$
Insufficient	$0 < A < k_c$
Sufficient	$A \geq k_c$

Deficit (unmet demand)

$$U_{i,c} = P_i \cdot \max\left(0,\ 1 - \frac{A_{i,c}}{k_c}\right)$$

A = facilities within 15 min; districts ranked by deficit.

Decisions

- ***implicit*:** a class per category, not one index: a planner sees what is missing
- **W7b:** combined score 0–100 over chosen categories, computed in the browser; never stored, ranked or used by the recommender
- With the defaults, one category gives $100\,(1 - U_{i,c}/P_i)$: map, deficit and recommender agree
- **M2 open:** transit class bounds still from Copenhagen

Source: docs/methodology.md §3, §4, §4b (W7b, v0.9), §7 (M2); Mouratidis (2026), *Travel Behaviour and Society* 42, 101151.

The recommender is greedy because the objective allows it: within 2.8 % of the exact optimum in every test

ID	Decision	Instead of	Why
§5	Gain = reduction of the deficit sum (ΔW)	Distance to the nearest facility	Same measure as the map; submodular, so greedy reaches $\geq 1 - 1/e$
V4	Greedy in the pipeline, exact MILP only as benchmark	MILP in every run	At most 2.8 % below the optimum (mean 0.17 %)
6 Nov	Candidates = 100 m cells in the urban boundary; industrial land kept; heritage areas not excluded	Exact parcels	Shops and doctors sit on commercial land; heritage protection excluded the Altstadt
W6	Cycling stays out of the recommender	–	Valhalla needs 4.0 GiB against a 2.5 GiB limit
§5b	Housing potential asks the reverse question; display only	–	Where would new homes already have good access?

Back-test V3: 80–98 % of picks lie within 500 m of a removed facility (random: 8–36 %).

Source: docs/methodology.md §5, §5b, change log v0.6; notes N12 (W6); validation report V3, V4 (run 20261106T142630Z).

One command refreshes, validates and publishes a run atomically; a failed run never reaches the app

acquire → **validate inputs** → tiles → count → score → housing → recommend → **validate results** → publish → cleanup

Pipeline

- **O19:** `make refresh` runs everything, incl. tile build and swap
- ***implicit*:** idempotent stages; each writes a manifest the next one checks
- ***implicit*:** a failed check stops the run before `publish`; we never loosen a check to pass (N16)
- ***implicit*:** each run gets its own DB schema; `current` switches only after success

Serving

- ***implicit*:** PostGIS, FastAPI, React + MapLibre; static web build behind Caddy
- **O9:** map data as GeoJSON (59 KB gzip per layer)
- **O1, O2:** one Hetzner server, $\geq$ 8 GB RAM, 20 € in total, free `sslip.io` hostname (plan to verify)
- Hosting changes are open (question 1)

Source: docs/architecture.md §2, §7, §8, §11, §13 (O1, O2, O9, O19); notes N16; implementation_plan.md (P6).

We work docs first: binding documents with change logs, an owner per folder and reviewed pull requests

Documents

- `architecture.md` (v0.23) and `methodology.md` (v0.11) win over code and plans
- Every change: new version and a change-log line with date, what and why
- Data contract between pipeline and API, tested in CI against PostGIS
- `AGENTS.md`: the same rules for coding agents

Team

- **T2:** one owner and one reviewer per folder
- Branch per topic, pull requests into `dev`; nobody approves their own
- **T1:** contributions table plus conventions (individual grading)
- ***implicit*:** one Python lockfile (uv) and Docker Compose for everyone

Source: AGENTS.md; CONTRIBUTING.md; docs/project_management_decisions.md (T1, T2, T8); docs/architecture.md §14.

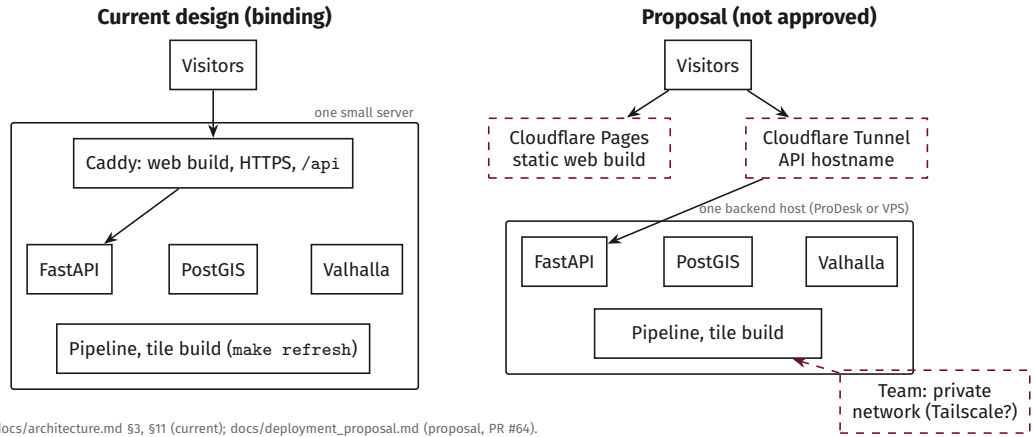
Four questions need your answer; for each we bring a recommendation

#	Question	Our recommendation	Blocks
1	Frontend on Cloudflare Pages, backend on one host?	Decide after your answers to 2 and 3	Deployment
2	Is the HP ProDesk acceptable as backend host, or is a rented VPS required?	Measure the first full refresh on the ProDesk, then decide	Server, budget
3	Is Moran's I and hot-spot analysis the "adequate method" (M5)?	Yes: optimization plus spatial statistics	Scope of V8
4	Which known limits do you accept, and which must we fix?	Accept and state; fix OSM gaps in OSM	Final report

Smaller points, including the scale of the project, follow after question 4.

Source: docs/deployment_proposal.md (proposal, PR #64); methodology.md §7 (M5); notes N11, N13, N16.

Question 1: should the frontend move to Cloudflare Pages? Socke's proposal is not approved



Source: docs/architecture.md §3, §11 (current); docs/deployment_proposal.md (proposal, PR #64).

Question 2: the ProDesk saves rent, a VPS saves operating effort; neither is chosen yet

	HP ProDesk (i5-8500T, 16 GB RAM)	Rented VPS ($\geq$ 8 GB, O1)
Benefit	Hardware is available; no rent	Data-centre power and network; remote management
To check	Power and network uptime, remote recovery, storage, backups, who maintains it	Monthly cost within the 20 € budget, disk size, provider limits, backups
Evidence needed	First full tile build and refresh measured on the host	Same measurement on the chosen plan

Please clarify: does “keep VPS and hosting separate” require a rented VPS? What uptime does the course demonstration need? Is a private network for team access (for example Tailscale) acceptable, and how should you reach the read-only database account?

Source: docs/deployment_proposal.md (proposal, PR #64); docs/architecture.md §11 (resource estimate to be verified on first deployment).

Question 3: Moran's I shows strong clusters between cells but none between districts; is this the adequate method?

What we did (V8)

- Global and local Moran's I on cells and on districts
- Cells: Moran's I 0.72–0.88, strong clusters
- Between districts: none

Our reading: where access is poor is a neighbourhood effect, not a district effect.

We ask

- Is optimization (recommender) plus spatial statistics enough for "traditional statistics or machine learning"?
- Or do you expect a fitted statistical or ML model?
- Should the app show the LISA map?

Source: validation report V8 (run 20261106T142630Z); methodology.md §7 (M5); notes N13.

Question 4: five known limits remain; we propose to state them openly and fix only the cheap ones

Limit	Effect	Proposal
Wheelchair on slopes	Steep sections are not avoided; hillside counts are optimistic (N11)	Accept, state on data page
OSM gaps	13 of 104 Kitas and 2 of 38 schools have no OSM match (V2)	Map them in OSM
Extended categories	No completeness check; monotonicity check at 10 of 1,699 cells, fails above 1 % (N16)	Mark as indicative
Cycling recommender off	12 zones would be added (W6)	Accept, document
Housing candidates	Land use not fully excluded; buildable not known (N15)	Decide after map check

Shares depend strongly on speed and budget (V5): we always name the profile and budget next to a share.

Source: notes N11, N12, N13, N15, N16; validation report V2, V5 (run 20261106T142630Z); run 20261110T233930Z.

Smaller points: the interim format, database access and whether a larger area is of interest

For the supervisors

- Interim format: length and level of results you expect
- Read-only database access: SSH tunnel as planned, or via the access model of question 1?
- Compute: may we use bwUniCluster for optional large runs?
- **Scale** (asked before 4 Nov): is one well-validated city enough for the course, and is the Germany outlook of interest?

Team-internal, still open

- O14: chat for the failure webhook
- O16: interim demo local (proposal) or on a short-lived server
- M1, M2: final category list and transit thresholds
- V7 and V9 need people (local knowledge; rerun from README)

Source: docs/project_management_decisions.md §7 (earlier supervisor questions); docs/architecture.md §13 (open); methodology.md §7.

Until 22 January we finish Tübingen: close the validations, deploy, and deliver

When	What
25 Nov	Interim presentation (`v0.1.0`); demo local unless decided otherwise (O16)
from ∼1 Dec	Server live after the hosting decision; weekly refresh by timer
Dec – Jan	V7 (local knowledge) and V9 (rerun from the README); M1 final categories, M2 transit classes; recommender decision cards and equity mode (M4); LISA map if wanted
10 – 15 Jan	Feature freeze: Stretch only if Core passes validation
19 Jan	Internal freeze: deployed, tested, V9 done
22 Jan	Submission (`v1.0.0`): code, hosted app, validation report, video, summary
28 / 29 Jan	Final slides / final presentation with live demo

Source: docs/features.md §9 (roadmap, cut order); project_management_decisions.md §8; notes N12, N13; methodology.md §7.

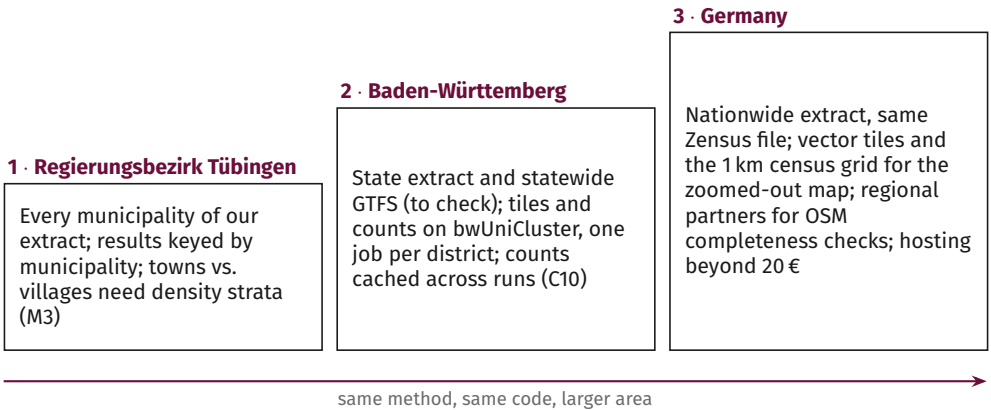
Our long-term goal is all of Germany, and the core choices already scale

Building block	Today: Tübingen	Germany
Residents	Zensus 100 m grid, filtered from the nationwide file	Same file: 3,088,037 populated cells, 82.6 million residents
Network, facilities	Geofabrik extract of the Regierungsbezirk Tübingen, clipped	Geofabrik extract of Germany, same format and code
Study area	One municipality key (AGS) in the config	All municipalities, one run each, in parallel
Routing	Valhalla tiles built on our server	Valhalla routes up to planet scale; build on bwUniCluster (§11.1)
Transit	NVBW GTFS (naldo)	A nationwide GTFS feed (to check, e.g. DELFI)

Vision of the team, not part of the course plan: a second region is out of scope for Core (architecture §12). Nothing on this or the next two slides is built.

Source: Zensus file of run 20261110T233930Z (grid sum, perturbed); config/config.yaml (`study_area`, `sources`); docs/architecture.md §4, §11.1, §12.

We would get there in three steps: the region we already download, Baden-Württemberg, then Germany



Source: docs/architecture.md §5.2 (1 km grid for coarser levels), §9 (vector tiles only if needed), §11.1; methodology.md §7 (M3); proposal of the team, not in the docs' plan.

Four problems decide whether Germany is feasible: compute, data quality, comparability and serving

Compute. `count` takes 254 s for 1,699 cells on a laptop. Germany has about 1,800× the cells: naively ∼5 days on one machine, so parallel jobs and caching.

Data quality. OSM completeness differs by region; V2 needs an official list per category and region; the extended categories are unchecked even in Tübingen.

Comparability. 15 minutes and k_c fit a city, not a village: density strata (M3) become necessary; profile and budget next to every share (V5).

Serving. Today one map layer is 59 KB of GeoJSON (gzip); Germany needs vector tiles, aggregated zoom levels and a server beyond the 20 € budget.

Source: notes N16 (count runtime, run 20261110T233930Z); naive linear estimate from the Zensus file; implementation_plan.md P6 (payload); methodology.md M3; validation report V2, V5.

With your answers we record the decisions, finish Tübingen and keep the design open for more

If you decide …	We do …
Pages and a backend host	Update `architecture.md`, plan production Compose, measure the first full refresh on the host
Moran's I is adequate	Finish V8 in the report; add the LISA map if you wish
Limits accepted	State them on the data page; map missing Kitas and schools in OSM
Interim format	Fit the 25 Nov deck and demo to it
A larger area is of interest	Record the Germany path as future work in `notes.md`; keep every value in the config

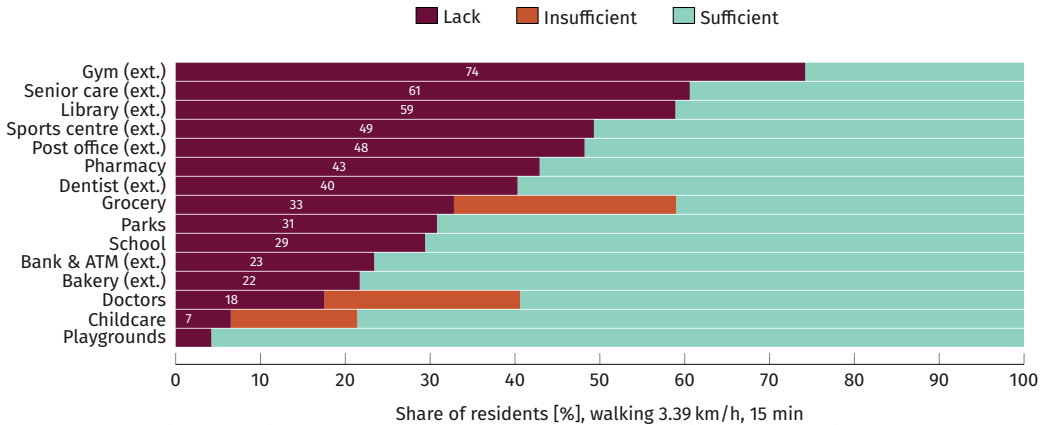
Open work stays open until done: V7, V9 and hosting are not finished.

Source: docs/features.md §9 (roadmap); docs/implementation_plan.md; AGENTS.md (no hard-coded values).

Thank you. Your decisions, please.

Hosting, the adequate method and the acceptable limits decide what we freeze next; all of Germany stays our long-term goal.

Backup: five of the eight extended categories are beyond a 15-minute walk for nearly half of the residents or more



Source: run 20261110T233930Z (OSM 8 Nov 2026), 90,355 residents in 1,699 cells. ext. = extended category, indicative (no completeness check).

Backup: cycling in the recommender, measured on 9 November

Variant	Runtime	Pipeline peak	Valhalla peak
Without cycling (3 profiles)	44 s	1.6 GB	1.2 GiB
+ cycling, 100 m candidates	295 s	2.0 GB	4.0 GiB
+ cycling, 200 m candidates	202 s	1.7 GB	3.9 GiB

Limit E6: runtime $\leq$ 10 min, Valhalla $\leq$ 2.5 GiB. Memory is set by the 3.75 km source sets, not by the number of candidates. The limit came from the 4 GB server; with the $\geq$ 8 GB target it can be set again (N12).

Source: notes N12 (W6), macOS/Docker Desktop.

Backup: validation experiments V1–V11

#	Experiment	Key result
V1	Matrix vs. isochrone polygons	Polygons add 18–20 % false pairs, 47 % across barriers
V2	OSM completeness	Childcare recall 88 %, precision 98 %; schools 95 % / 95 %
V3	Recommender back-test	80–98 % of picks within 500 m of a removed facility (only the removed deficit counts)
V4	Greedy vs. exact MILP	At most 2.8 % below the optimum (mean 0.17 %)
V5	Sensitivity (k, speed, budget)	Shares move strongly; district rankings stable (median Spearman 0.78–0.95)
V6	Centroid vs. building origin	1.9 % (walking) / 2.8 % (wheelchair) change class
V7	Local knowledge	Not yet run
V8	Spatial statistics	Moran's I 0.72–0.88 on cells; none between districts
V9	Reproducibility	Not yet run
V10	Matrix vs. single routes	Median $\lvert\Delta\rvert$ 0.5 s; 0 of 900 false negatives
V11	With vs. without slopes	3.3–3.6 % change class, mostly on hillsides

Source: docs/validation_report.md, run 20261106T142630Z (Core categories of that run, parks and playgrounds still one category); the extended categories are not validated yet.